Shuo Lv

AI Agents · AI for Science · Cognitive Neuroscience · Complex Networks · Pattern Recognition

Email · Beijing, China · Homepage · GitHub · Scholar · ORCID

Education

- **Beijing Institute of Technology** — Beijing, China
  M.Eng. in Electronic Information — *Sep. 2024 – Present*
- **Southwest University** — Chongqing, China
  B.Eng. in Data Science and Big Data Technology — *Sep. 2020 – Jun. 2024*
  Ranked **2/103**; received the **National Scholarship twice**, the Gratitude to Modern Chinese Scientists Scholarship, and the Top-Tier Scholarship.

Major Courses: Machine Learning; Mathematical Modeling; Optimization; Matrix Theory; Discrete Mathematics; Big Data Applications

Selected Publications

- **Lv, S.**, Li, J., Yang, R. *et al.* Three parsimonious spatiotemporal patterns in cerebellum reveal individual traits in function and behavior. *Nat Commun* **17**, 6232 (2026).
- **Lv, S.**, Wang, Y., Guo, C. & Zhang, L. Effects of experts on the coupling dynamics of complex contagion of awareness and epidemic spreading. *Nonlinear Dyn* **112**, 2367–2380 (2024).
- Yang, R., Wu, X., **Lv, S.** *et al.* Alterations in functional brain network topography in autism revealed by normative models. *Transl Psychiatry* (accepted Sep. 2026).
- Gao, T., Xue, M., Zheng, H., **Lv, S.** *et al.* BrainLMM: A Label-Free Framework for Mapping Multi-Semantic Representation in the Human Visual Cortex. *Proc. AAAI Conf. Artif. Intell.* **40**, 4176–4184 (2026).
- Guo, C., Wang, Y., **Lv, S.** *et al.* Impact of Sampling Disease Detection and Collective Intervention Measures in Epidemic Transmission. In *Proc. 20th Chinese Conf. on Computer Supported Cooperative Work and Social Computing (ChineseCSCW 2025)* (2025).
- Li, J., Zhang, Y., Wu, X. *et al.* Development of areal-level individualized homologous functional parcellations in youth. *Commun Biol* **8**, 1083 (2025).

Research Internship

ByteDance, Seed Foundation Model – AI4Math – *Research Intern, AI Agent Development* — *Nov. 2025 – May 2026*

Built a test-time-scaling multi-agent system on the Seed 2.0 model family for olympiad mathematics, designing Planner, Solver, and Verifier roles for strategy generation, parallel solving, proof verification, and backtracking; completed **5 iterations** over **100+ problems** and averaged **6.4/7** on an IMO problem set. Led an automated Lean 4 proof-QA workflow that produced **600+ formalized data items** and improved expert review efficiency by **70%+**. Standardized evaluation-data SOPs, delivering **2,000+ data items** and **11** production and quality specifications.

Research Experience

General-Purpose Cybersecurity Agent – *Lead Researcher* — *Jun. 2026 – Present*

Advisor: Libo Zhang, Southwest University. Built **5 specialized agents** and a **16-node bounded state machine** with LangGraph, separating LLM interpretation, planning, and reflection from explicit permissions, routing, and termination. Designed R0–R3 risk levels, Finding–Evidence validation, human approval for high-risk actions, and a two-layer state/domain blackboard for traceable decisions across code audit, penetration testing, incident response, and reverse engineering.

Low-Dimensional Spatiotemporal Dynamics of the Human Brain – *Lead Researcher* — *Dec. 2023 – Present*

Advisor: Guoyuan Yang, Beijing Institute of Technology. Applied complex principal component analysis to resting-state cerebellar fMRI and identified three dominant sparse patterns spanning zero-lag spatial structure and time-lag dynamics. Used sCCA and linear SVMs to associate individualized cerebellar activity with age, sex, cognition, and emotion, resulting in **2 articles**: a first-author *Nature Communications* article and one accepted by *Translational Psychiatry*.

Complex Social Awareness Modeling – *Lead Researcher* — *Apr. 2022 – Nov. 2024*

Advisor: Libo Zhang, Southwest University. Modeled expert influence and isolation behavior in epidemic prevention using prospect and regret theories. Random-network simulations quantified intervention benefits and revealed phase transitions in social awareness, resulting in **2 papers**: a first-author *Nonlinear Dynamics* article and a CCF Class C conference paper.

Intelligent Factory Security – *Lead Researcher* — *Apr. 2022 – Sep. 2023*

Advisor: Mingyang Zhong, Southwest University. Developed adaptive multimodal identity recognition for faces under abnormal illumination or occlusion and voices in noisy environments; used three-way decisions (accept, reject, defer) to exploit uncertainty estimates from visual and audio modalities.

Large Language Model for Early Childhood Education – *Researcher* — *Dec. 2023 – Oct. 2024*

Advisor: Libo Zhang, Southwest University. Built a personalized multimodal teaching-material generation system on vivo's BlueLM, integrating content generation, interactive dialogue, and AI-assisted drawing.

Xiangqi Game-Playing System – *Lead Developer* — *Apr. 2021 – Aug. 2021*

Advisor: Shenglin Li, Southwest University. Implemented a Java Chinese chess engine using pruned α–β search, custom search and evaluation functions, human-vs-computer play, and game-record storage.

Awards and Honors

Jun. 2023	China Robot and Artificial Intelligence Competition	**National First Prize**
Oct. 2024	China Collegiate Computing Contest – AIGC Innovation Track	**National First Prize**
Oct. 2023	Contemporary Undergraduate Mathematical Contest in Modeling	**National Second Prize**
Aug. 2022	Chinese Computer Game Championship	**National Second Prize**
Sep. 2024	China International College Students' Innovation Competition	**National Third Prize**

Academic Service

Teaching Assistant: Algorithm Design and Analysis; Introduction to Algorithms; Big Data Analytics (SWU).
Reviewer: *Nonlinear Dynamics*; *Brain Structure and Function*.

Technical Skills

Languages	Python, C++, Java, R, Lean 4, JavaScript, SQL, HTML, Bash
Software Stack	FastAPI, Pydantic, LangGraph, Guardrails, React, PyTorch
Tools	Git, Model Context Protocol (MCP), PE, FSL, Connectome Workbench, MATLAB, Adobe Photoshop, Premiere Pro, and InDesign